Refer to the **insert** for the list of pseudocode functions and operators.

1 A user requires a program that implements complex data encryption to a recognised international standard. A programmer has started the design of the program.

(a) The programmer decides to use a library routine to provide part of the solution. One reason for this is that library routines may perform functions that the programmer is unable to program themselves.

State **three** other benefits of using library routines in the development of the program.

1 .....

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2 .....

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3 .....

.....

[3]

(b) The programmer has identified a different part of the algorithm that would be appropriate to implement as a subroutine (a procedure or a function); no suitable library routine exists for this part.

(i) State **two** reasons why the programmer may decide to use a subroutine.

1 .....

.....

2 .....

.....

[2]

2 **(a)** A program contains a global 1D array of type `STRING` containing 65 elements.

An existing text file is used to store the data in the array. Only non-blank elements (those that do **not** contain an empty string) are written to the text file.

An algorithm will:
- write the first element of the array as a new line in the text file
- continue until all elements have been written, each to a new line of the text file.

Stepwise refinement is applied to the algorithm.

Describe up to **six** steps for this algorithm that could be used to produce pseudocode.

Do **not** use pseudocode statements in your answer.

Step 1 .....

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Step 2 .....

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Step 3 .....

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Step 4 .....

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Step 5 .....

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Step 6 .....

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[6]

(b) Iteration is one programming construct.

Identify **one** other programming construct that will be required when the algorithm from part **(a)** is converted into pseudocode and explain how it is used.

Construct .....

.....

Use .....

.....

[2]

2 `Data` is a global 1D array containing 20 elements of type `REAL`

An algorithm will:

- input a sequence of real values, one at a time
- assign each value to consecutive array elements, starting from index 1
- end when the value 99.9 is input, or all 20 elements have been assigned (the value 99.9 must **not** be stored in the array).

Complete the program flowchart to represent the algorithm:



[5]

3 A text file `OldFile.txt` contains IDs, names and email addresses for members of a club. Three information items are stored for each member and each item is stored on a separate line.

The example shows the information for the first two members in the first six lines of the file:

Line in file	Information item	Example data
1	member 1 ID	`"AB1234"`
2	member 1 name	`"Freddie Jones"`
3	member 1 email address	`"FreddieJ909@Cambridge.org"`
4	member 2 ID	`"BC2345"`
5	member 2 name	`"Sue Smith"`
6	member 2 email address	`"Sue1024@Cambridge.org"`

The member ID string is always two letters followed by four digits.

The file design is to be changed so that information for each user is stored as a single line of the file, with the character `'\'` used as a separator between data items.

For example, the single line for member 1 will be:

```
"AB1234\Freddie Jones\FreddieJ909@Cambridge.org"
```

An algorithm will produce a new file `NewFile.txt` from the contents of `OldFile.txt`

Assume:

- `LineX`, `LineY` and `LineZ` are of type `STRING` and are used to store the three items of information for each member
- `NewString` is a temporary variable of type `STRING`
- `OldFile.txt` exists and contains valid data.

(a) The algorithm to create the new file is expressed in steps.

Complete the following numbered steps:

Step 1 : open the file ..... in .....

Step 2 : open the file ..... in .....

Step 3 : read a line from ..... and store in .....

Step 4 : read a line from ..... and store in .....

Step 5 : read a line from ..... and store in .....

Step 6 : set `NewString` to .....

Step 7 : write `NewString` to .....

Step 8 : repeat from step ..... until .....

Step 9 : close both files.

[6]

4 A quiz has nine questions. There are:

- five easy questions each worth 3 points
- four hard questions each worth 5 points.

At the end of the quiz the points for each correctly answered question are added to give a total score.

A check is made to test that the total score is valid using a 1D array `CheckTotal` of type `BOOLEAN`. Each index value of the array represents a possible total score. The corresponding element value is `TRUE` if the index value represents a valid total score and `FALSE` otherwise.

The first nine rows of the array are:

Index value	Element value	Comment
0	`TRUE`	valid total score (no correct answers)
1	`FALSE`	invalid total score
2	`FALSE`	invalid total score
3	`TRUE`	valid total score (one 3-point question correct)
4	`FALSE`	invalid total score
5	`TRUE`	valid total score (one 5-point question correct)
6	`TRUE`	valid total score (two 3-point questions correct)
7	`FALSE`	invalid total score
8	`TRUE`	valid total score (one 3-point question and one 5-point question correct)

For example, a total score of 6 points is valid; the value of the array element at index value 6 is `TRUE`

(a) Write pseudocode to declare `CheckTotal` and to set all elements of the array to `FALSE`

All variables used must be declared.

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..... [4]

(c) The array elements have been assigned the required values.

A module `ValidateScore()` will take an integer value representing a total score as a parameter. It will return `TRUE` if the total score is valid, or `FALSE` if it is **not** valid.

Describe the algorithm for `ValidateScore()` using **four** steps.

Do **not** use pseudocode in your answer.

Step 1 .....

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Step 2 .....

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Step 3 .....

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Step 4 .....

.....

[4]

2 `Data` is a global 1D array containing 30 elements of type `STRING`

An algorithm will output:

- all non-blank elements (elements that do **not** contain an empty string)
- the final total of the number of elements output.

Complete the program flowchart to represent the algorithm:





[5]

5 A program is being designed in pseudocode.

The program contains the following declaration for the global array `MyData`:

```
DECLARE MyData : ARRAY[1:10000] OF STRING
```

A function `FindFirst()` is written to search the array for a given string and to return the index of the first element where that string is found, or to return –1 if the string is **not** found.

The function is written in pseudocode as shown:

```
FUNCTION FindFirst(SearchString : STRING) RETURNS INTEGER
   DECLARE Index, FoundAt : INTEGER
   FoundAt ← -1
   FOR Index ← 1 TO 10000
      IF MyData[Index] = SearchString THEN // outer conditional clause
         IF FoundAt = -1 THEN              // inner conditional clause
            FoundAt ← Index
         ENDIF
      ENDIF
   NEXT Index
   RETURN FoundAt
ENDFUNCTION
```

(a) (i) Comment on why the programmer chose –1 as the initial value of `FoundAt`

.....

..... [1]

(iii) The **inner** conditional clause ensures that only the index of the **first** matching element, if any, is returned.

Explain how this clause works.

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..... [1]

6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	4 + 1 + 6 + 2 = 13
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

(a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number .....

Explanation .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is being developed for the management of a sports centre.

(a) The programmer developing the software decides to use modules (procedures or functions) and local variables. They also decide not to use global variables.

(i) State **two** reasons why the programmer decides to use modules.

1 .....

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2 .....

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[2]

4 A programmer is developing a new stock control program for a shop owner to produce sales reports.

(a) A text file `Sales.txt` stores strings representing the number of items sold.

The file contains 52 lines, each representing the number of items sold in each week of the year.

For example:

File line number	File data
1	"350"
2	"502"
3	"434"
49	"492"
50	"872"
51	"772"
52	"201"

The example shows that 502 items were sold in week 2.

The owner requires a module to output all week numbers where the number of items sold is greater than 500.

Describe an algorithm that produces the required module.

Do **not** include pseudocode statements in your answer.

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..... [5]

2 A program to calculate the pay of employees working for a company is being designed.

(a) Stepwise refinement has been used in the design of the program.

Describe stepwise refinement.

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..... [2]

3 An algorithm is designed to generate and output two unique random integers. Each integer value is between –10 and 10 inclusive.

If both integers output are negative, a third random integer between 30 and 35 inclusive will be generated and output.

Write pseudocode for this algorithm.

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4 An algorithm will:

- input 100 integer values, one value at a time
- store the first value input into the first location of the array `Number`
- store the next input value in the next unused location of the array `Number`
- output the contents of `Number` array in the opposite sequence to that in which the values were input.

Complete the program flowchart to represent the algorithm.

Variable declarations are **not** required.





[6]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is being developed to help the manager of a shop control the stock.

(a) An identifier table has been used during the design stage.

Complete the identifier table:

Example value	Explanation	Variable name	Data type
`"Fruit"`	a category of stock that is sold in the shop		
`20/02/2025`	when an item was sold		
`12.67`	the cost of an item		
`TRUE`	to indicate if an item is in stock		

[4]

(b) A module `Sales()` is part of the stock control program.

The table contains pseudocode extracts from the module `Sales()`

Each extract may include all or part of:

- assignment
- selection
- iteration (repetition).

Complete the table by placing one or more ticks (✓) in each row:

Pseudocode extract	Assignment	Selection	Iteration
`Result ← CalculateTotal()`			
`WHILE IsClosed`			

`REPEAT`
`  INPUT Value`
`UNTIL Sales[4] > Value`			

`IF Sales[Current] <= 150 THEN`
`  Discount ← TRUE`
`ENDIF`			

`CASE OF Option`			

[5]

(c) Decomposition has been used to design the program to help the shop manager control the stock.

Describe decomposition.

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..... [3]

3 A student has been asked to create a simple guessing game program. This program will generate a random integer value between 1 and 100. It will then repeatedly prompt the user to input an integer value until they input the randomly generated value.

The student has written a structured English description:

step 1 – randomly generate an integer value between 1 and 100 inclusive
step 2 – prompt the user to input an integer value
step 3 – output an appropriate message if the value input was too high; then repeat from **step 2**
step 4 – output an appropriate message if the value input was too low; then repeat from **step 2**
step 5 – output an appropriate message if the value input was the same value that was randomly generated; then end the program.

Write a pseudocode algorithm from this structured English description.

Assume no input validation is needed.

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..... [8]

1 (a) The following table contains pseudocode examples.

Each example may contain statements that relate to one or more of the following:

- selection
- iteration (repetition)
- subroutine (procedure or function).

Complete the table by placing **one or more** ticks ('✓') in each row.

Pseudocode example	Selection	Iteration	Subroutine

`FOR Index ← 1 TO 3`
`  IF Safe[Index] = TRUE THEN`
`  Flap[Index] ← 0`
`  ENDIF`
`NEXT Index`			

`CASE OF Compound(3)`			
`REPEAT UNTIL AllDone() = TRUE`			
`WHILE Result[3] <> FALSE`			

[4]

2 An algorithm will:

1. prompt and input a sequence of 100 integer values, one at a time
2. sum the positive integers
3. output the result of the sum.

(a) Write pseudocode for the algorithm.

Assume the value zero is neither positive nor negative.

You must declare all variables used in the algorithm.

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..... [5]

(b) The algorithm requires the use of basic constructs. One of these is sequence.

Identify **one other** basic construct required by the algorithm **and** describe how it is used.

Construct .....

Use .....

.....

[2]

3 The implementation of a linked list uses an integer variable and a 1D array `List` of type `Node`.

Record type `Node` is declared in pseudocode as follows:

```
TYPE Node
   DECLARE Data : STRING
   DECLARE Pointer : INTEGER
ENDTYPE
```

The array `List` is declared in pseudocode as follows:

```
DECLARE List : ARRAY[1:200] OF Node
```

The linked list will operate as follows:

- Integer variable `HeadPointer` will store the array index for the first node in the linked list.
- The `Pointer` field of a node contains the index value of the next array element (the next node) in the linked list.
- The value 0 is used as a null pointer.

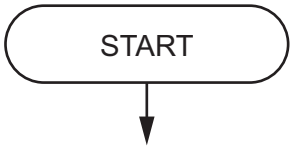
(a) (i) State why the value 0 has been selected as the null pointer.

.....

..... [1]

(b) The array `List` will be initialised so that each node points to the following node. The last node will contain a null pointer.

Complete the program flowchart to represent the algorithm for this operation.





[4]

(c) An algorithm outputs the `Data` field from all nodes in the array `List`. The order the `Data` is output should be the same order it is stored in the linked list.

Describe the algorithm in **four** steps.

Do **not** use pseudocode statements in your answer.

Step 1 .....

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Step 2 .....

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Step 3 .....

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Step 4 .....

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[4]

7 A coffee shop owner wants to introduce a computerised loyalty card system.

A programmer discusses the details of the system with the shop owner.

(a) Identify the stage of the program development life cycle that this discussion is part of **and** give a document that will be produced during this stage.

Stage .....

Document .....

.....

[2]

(b) The shop will give each customer a loyalty card that displays a unique customer ID as a bar code. A customer will be able to present their card each time they make a purchase. The system will scan the bar code, calculate points, and add them to the customer's total. When the customer next makes a purchase and presents their card, they will have the option to exchange points for a discount.

The designer decides that this activity will be handled by a new module. Decomposition will be used to break the problem of designing the new module down into sub-problems (sub-modules).

Identify **four** sub-modules that could be used in the design of the new module **and** describe their use.

Sub-module 1 .....

Use .....

.....

Sub-module 2 .....

Use .....

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Sub-module 3 .....

Use .....

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Sub-module 4 .....

Use .....

.....

[4]

6 A shop sells sandwiches and snacks. The owner chooses a 'daily special' sandwich which is displayed on a board outside the shop. Each 'daily special' has two different fillings and is made with one type of bread.

The owner wants a program to randomly choose the 'daily special' sandwich.

The program designer decides to store the possible sandwich fillings in a 1D array of type string.

The array is declared in pseudocode as follows:

```
DECLARE Filling : ARRAY [1:35] OF STRING
```

Each element contains the name of one filling.

An example of the first five elements is as follows:

Index	Element value
1	`"Cheese"`
2	`"Onion"`
3	`"Salmon"`
4	`"Anchovies"`
5	`"Peanut Butter"`

A second 1D array stores the possible bread used:

```
DECLARE Bread : ARRAY [1:10] OF STRING
```

Each element contains the name of one type of bread.

An example of the first three elements is as follows:

Index	Element value
1	`"White"`
2	`"Brown"`
3	`"Pitta"`

Both arrays may contain unused elements. The value of these will be an empty string and they may occur anywhere in each array.

A procedure `Special()` will output a message giving the 'daily special' sandwich made from **two** randomly selected different fillings and **one** randomly selected bread.

Unused array elements must **not** be used when creating the 'daily special' sandwich.

Using the above examples, the output could be:

```
"The daily special is Cheese and Onion on Brown bread."
```

(a) Complete the pseudocode for the procedure `Special()`.

Assume that both arrays are global.

```
PROCEDURE Special()
```

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```
ENDPROCEDURE
```

[7]

(b) The owner decides that some combinations of fillings do not go well together. For example, anchovies and peanut butter.

Describe how the design could be changed to prevent certain combinations being selected.

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..... [2]

7 A coffee shop runs a computerised loyalty card system.

Customers are issued with a loyalty card with their name together with a unique customer ID.

Loyalty points are added to their card each time they spend money at the shop.

The following information is stored for each customer: ID, name, home address, email address, mobile phone number, date of birth, number of points, date of last visit and amount of money spent at last visit.

A new module will generate a personalised email message to each loyalty card customer who has not visited the coffee shop in the last three months. The message will include a unique voucher code which can be used to authorise a discount if the customer goes to the shop within the next two weeks.

(a) Identify **three** items of customer information that will be used by the new module **and** justify your choices.

Item 1 .....

Justification .....

.....

Item 2 .....

Justification .....

.....

Item 3 .....

Justification .....

.....

[3]

(b) Identify the computational thinking skill that you needed to use to answer part **(a)**.

..... [1]

BLANK PAGE

3 An algorithm will output the **last** three lines from a text file `Result.txt`

The lines need to be output in the same order as they appear in the file.

Assume:

- Three variables `LineX`, `LineY` and `LineZ` will store the three lines. These are of type string and all three variables have been initialised to an empty string.
- The file exists and contains **at least** three lines.

(a) The algorithm to output the lines is expressed in eight steps.

Complete the steps.

1. Open the file .....
2. Loop until .....
3. ..... and store in `ThisLine`
4. Assign `LineY` to `LineX`
5. Assign `LineZ` to `LineY`
6. Assign `ThisLine` to `LineZ`
7. After the loop, ...
8. Output `LineX`, `LineY`, `LineZ`

[4]

(b) Explain the purpose of steps 4, 5 and 6 in the algorithm from part **(a)**.

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..... [1]

4 A program includes the following assignment statement:

```
Result ← STR_TO_NUM(x) / STR_TO_NUM(y)
```

When the program evaluates the expression in the statement, it performs a calculation.

Variable `Result` is of type real and variables `x` and `y` are of type string.

Two checks are required before the calculation is performed:

1. The two strings represent valid numeric values.
2. The numeric value of string `y` is not zero.

(a) Identify the type of error that could occur if these checks are **not** carried out **and** state a cause of this error.

Type .....

.....

Cause .....

.....

[2]

(b) The designer considers implementing the **checks and calculation** as a module (a procedure or a function). One reason for this is that the same checks and calculations are performed at several places in the program.

Give **another** reason why this is a suitable approach **and** state what is avoided by this approach.

Reason .....

.....

Avoided .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program uses many complex algorithms.

One algorithm is repeated in several places. The code for the algorithm is the same wherever it is used, but the calculations within the algorithm may operate on different data.

The result of each calculation is used by the code that follows it.

It is decided to modify the program and implement the algorithm as a separate module.

(a) (i) State **two** benefits of this modification to the existing program.

1 .....

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2 .....

.....

[2]

2 **(a)** A program uses a global 1D array of type string and a text file.

An algorithm that forms part of the program is expressed as follows:
- copy the first line from the file into the first element of the array
- copy the second line from the file into the second element of the array
- continue until all lines in the file have been copied into the array.

Stepwise refinement is applied to the algorithm.

Outline **five** steps for this algorithm that could be used to produce pseudocode.

Assume there are more elements in the array than lines in the file.

Do **not** use pseudocode statements in your answer.

Step 1 .....

.....

Step 2 .....

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Step 3 .....

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Step 4 .....

.....

Step 5 .....

.....

[5]

(b) Sequence is one programming construct.

Identify **one other** programming construct that will be required when the algorithm from part **(a)** is converted into pseudocode **and** explain its use.

Construct .....

.....

Use .....

.....

[2]

BLANK PAGE

3 A record structure is declared to hold data relating to components being produced in a factory:

```
TYPE Component
   DECLARE Item_ID : STRING
   DECLARE Reject : BOOLEAN
   DECLARE Weight : REAL
ENDTYPE
```

The factory normally produces a batch (or set) of 1000 components at a time. A global array is declared to store 1000 records for a batch:

```
DECLARE Batch : ARRAY [1:1000] OF Component
```

Two global variables contain the minimum and maximum acceptable weight for each component. The values represent an inclusive range and are declared as:

```
DECLARE Min, Max : REAL
```

(a) (i) A program uses a variable `ThisIndex` as the array index to access a record.

Write a pseudocode clause to check whether or **not** the weight of an individual component is within the acceptable range.

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..... [3]

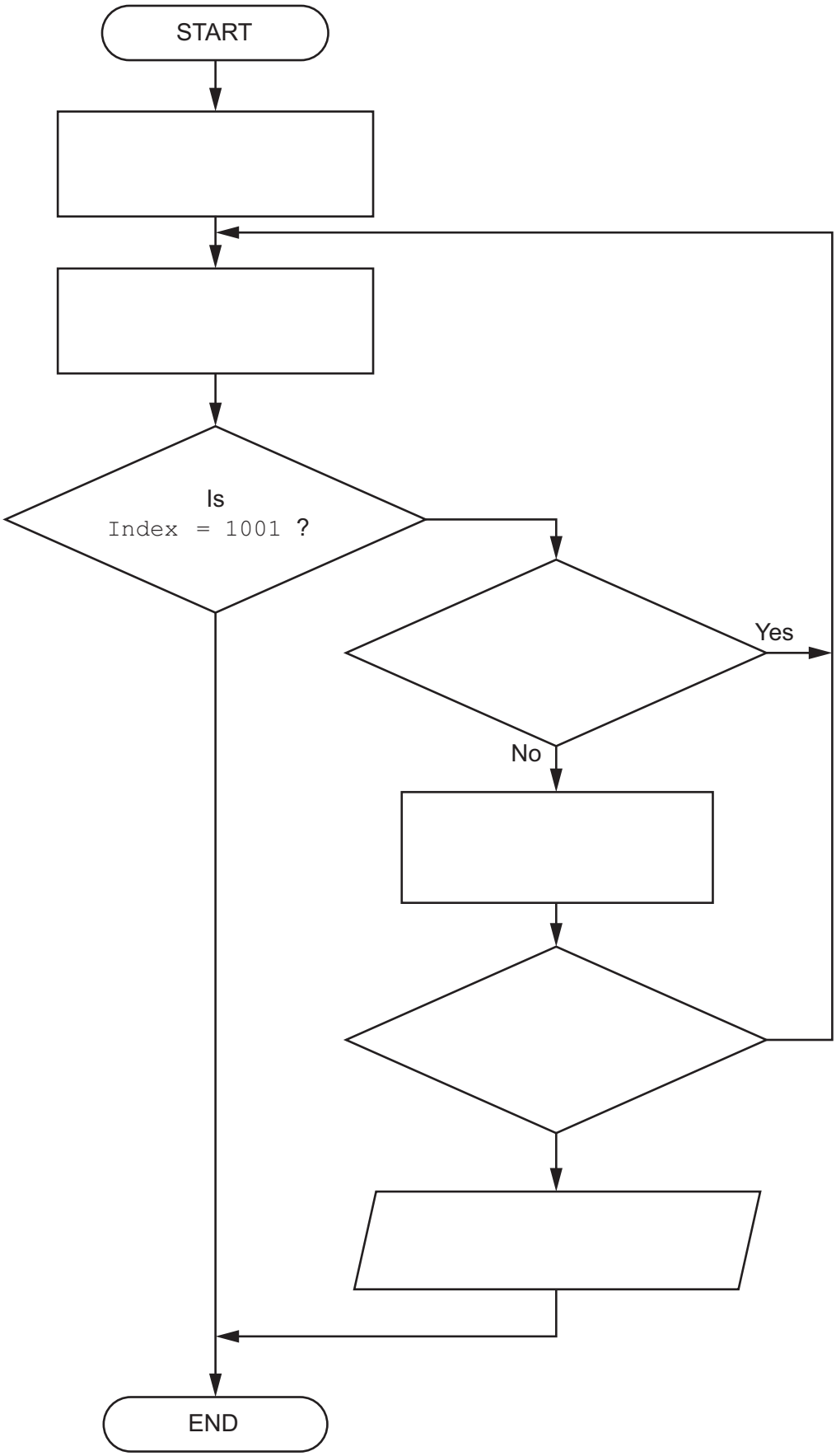
(b) A module `InRange()` will:

- be called with an integer parameter representing an index value of a record in the `Batch` array
- check if the weight of the indexed component is within the acceptable range
- return `TRUE` if the weight is in the range and `FALSE` if it is **not**.

A module `BatchCheck()` will:

- iterate through a batch of 1000 component records
- call module `InRange()` to check each individual component record
- keep a count of the number of components that fail
- output a suitable warning message and immediately stop if the number of failed components exceeds 5.

Complete the program flowchart to represent the algorithm for module `BatchCheck()`.



[5]

4 A procedure `TwoParts()` will input a sequence of real values, one at a time.

The procedure will:
- process the sequence in two parts
- form a first total by adding the values until the first zero
- form a second total by adding the values after the first zero until the second zero
- output the average of the two totals, together with a suitable message.

Values input in the first part are totalled using global variable `TotalA` and those input in the second part are totalled using global variable `TotalB`.

(a) Write pseudocode for the procedure `TwoParts()`.

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..... [6]

Refer to the **insert** for the list of pseudocode functions and operators.

1 (a) The following table contains pseudocode examples.

Each example may contain statements that relate to one or more of the following:
- selection
- iteration (repetition)
- input/output.

Complete the table by placing **one or more** ticks (✓) in each row.

Pseudocode example	Selection	Iteration	Input/Output

`FOR Index ← 1 TO 10`
`  Data[Index] ← 0`
`NEXT Index`			

`WRITEFILE ThisFile, "****"`			
`UNTIL Level > 25`			

`IF Mark > 74 THEN`
`  READFILE OldFile, Data`
`ENDIF`			

[4]

(b) Program variables have data types as follows:

Variable	Data type
`MyChar`	`CHAR`
`MyString`	`STRING`
`MyInt`	`INTEGER`

Complete the table by filling in each gap with a function (from the **insert**) so that each expression is valid.

Expression
`MyInt ← ..... (3.1415926)`
`MyChar ← ..... ("Elwood", 3, 1)`
`MyString ← ..... ( ..... (27.509))`
`MyInt ← ..... ( ..... ("ABC123", 3))`

[4]

(c) The variables given in part **(b)** are chosen during the design stage of the program development life cycle.

The choices are to be documented to simplify program maintenance.

State a suitable way of documenting the variables **and** give **one** piece of information that should be recorded, in addition to the data type.

.....

.....

..... [2]

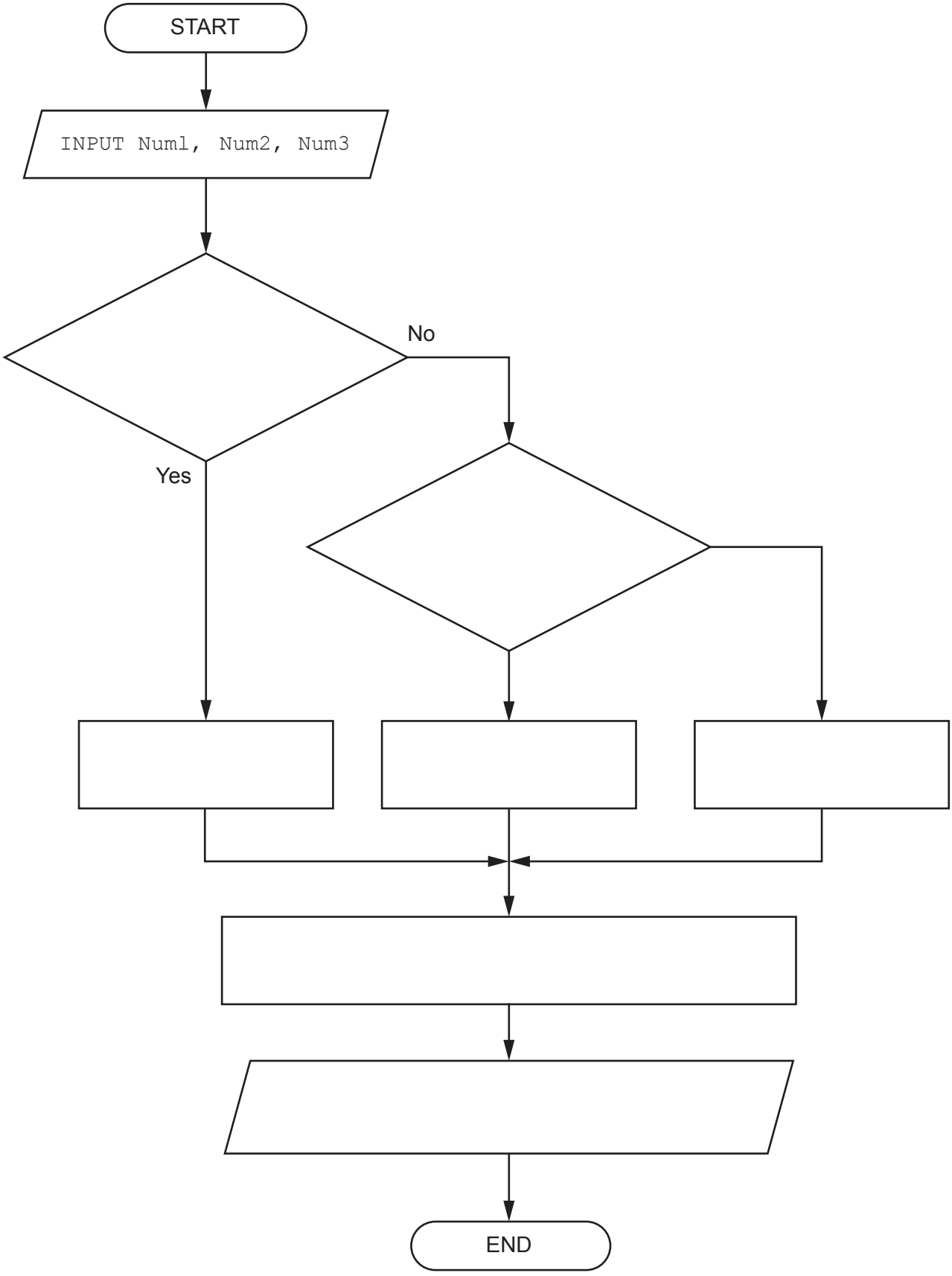
2 A program is being developed.

(a) An algorithm for part of the program will:

- input three numeric values and assign them to identifiers `Num1`, `Num2` and `Num3`
- assign the largest value to variable `Ans`
- output a message giving the largest value and the average of the three numeric values.

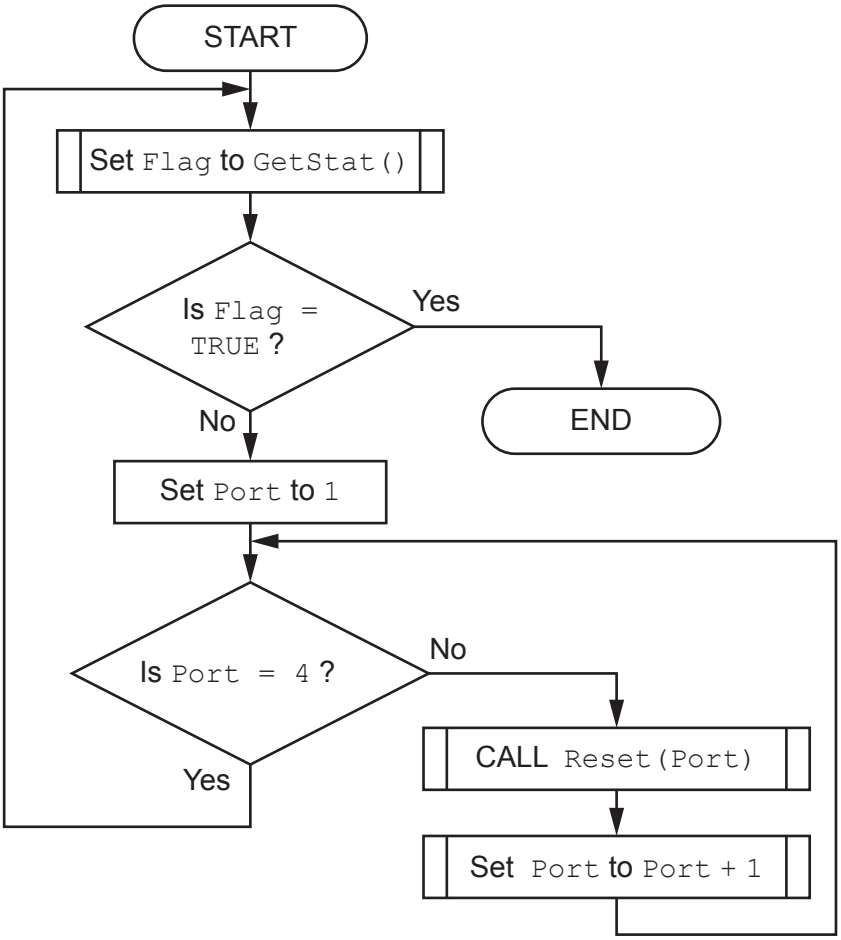
Assume the values are all different and are input in no particular order.

Complete the program flowchart on page 5 to represent the algorithm.



[5]

(b) A different part of the program contains an algorithm represented by the following program flowchart:



Write pseudocode for the algorithm.

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..... [5]

7 A fitness club has a computerised membership system.

The system stores information for each club member: name, home address, email address, mobile phone number, date of birth and exercise preferences.

Many classes are full, and the club creates a waiting list for each class. The club adds details of members who want to join a class that is full to the waiting list for that class.

When the system identifies that a space is available in one of the classes, a new module will send a text message to each member who is on the waiting list.

(a) Decomposition will be used to break the new module into sub-modules (sub-problems).

Identify **three** sub-modules that could be used in the design **and** describe their use.

Sub-module 1 .....

Use .....

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Sub-module 2 .....

Use .....

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Sub-module 3 .....

Use .....

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[3]

8 A teacher is designing a program to process pseudocode projects written by her students.

Each student project is stored in a text file.

The process is split into a number of stages. Each stage performs a different task and creates a new file.

For example:

File name	Comment
`MichaelAday_src.txt`	Student project file produced by student Michael Aday
`MichaelAday_S1.txt`	File produced by stage 1
`MichaelAday_S2.txt`	File produced by stage 2

(a) Suggest a reason why the teacher's program has been split into a number of stages **and** give the benefit of producing a different file from each stage.

Reason .....

.....

Benefit .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 An algorithm is developed in pseudocode before being coded in a programming language.

(a) The following table shows four valid pseudocode assignment statements.

Complete the table by giving an appropriate data type to declare each of the variables `A`, `B`, `C` and `D`.

Assignment statement	Data type
`A ← LEFT(MyName, 1)`	
`B ← Total * 2`	
`C ← INT(ItemCost) / 3`	
`D ← "Odd OR Even"`	

[4]

(c) The variable names `A`, `B`, `C` and `D` in part **(a)** are **not** good programming practice.

(i) State why these variable names are **not** suitable.

.....

..... [1]

(ii) Identify **one** problem that these variable names might cause.

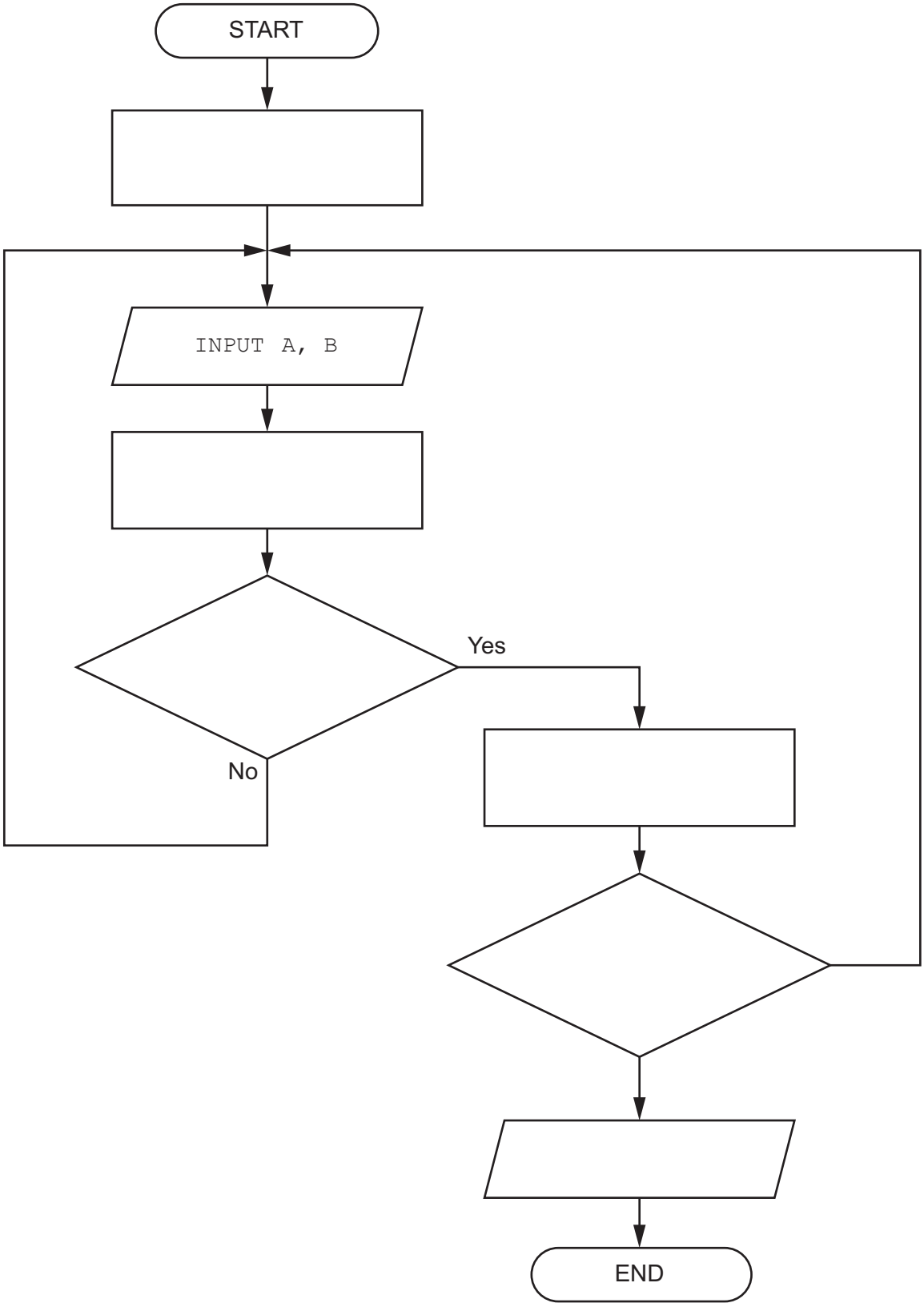
.....

..... [1]

2 An algorithm has three steps. It will:

1. repeatedly input a pair of numeric values `A` and `B`
2. count the number of pairs that are input until `A` has been greater than `B` 10 times
3. output the number of pairs that were input.

(a) Complete the program flowchart.



[5]

7 A fitness club has a computerised membership system. The fitness club offers a number of different exercise classes.

The following information is stored for each club member: name, home address, email address, mobile phone number, date of birth and the exercise(s) they are interested in.

(a) When an exercise class is planned, a new module will send personalised text messages to each member who has expressed an interest in that exercise. Members wishing to join the class send a text message back. Members may decide **not** to receive future text messages by replying with the message 'STOP'.

The process of abstraction is used to filter out unnecessary information.

(i) State **one** advantage of applying abstraction to this problem.

.....

..... [1]

(ii) Identify **three** items of information that will be required by the new module.
Justify your choices with reference to the given scenario.

Item 1 required .....

Justification .....

.....

Item 2 required .....

Justification .....

.....

Item 3 required .....

Justification .....

.....

[3]

(iii) Identify **two** operations that would be required to process data when the new module receives a text message back from a member.

Operation 1 .....

.....

Operation 2 .....

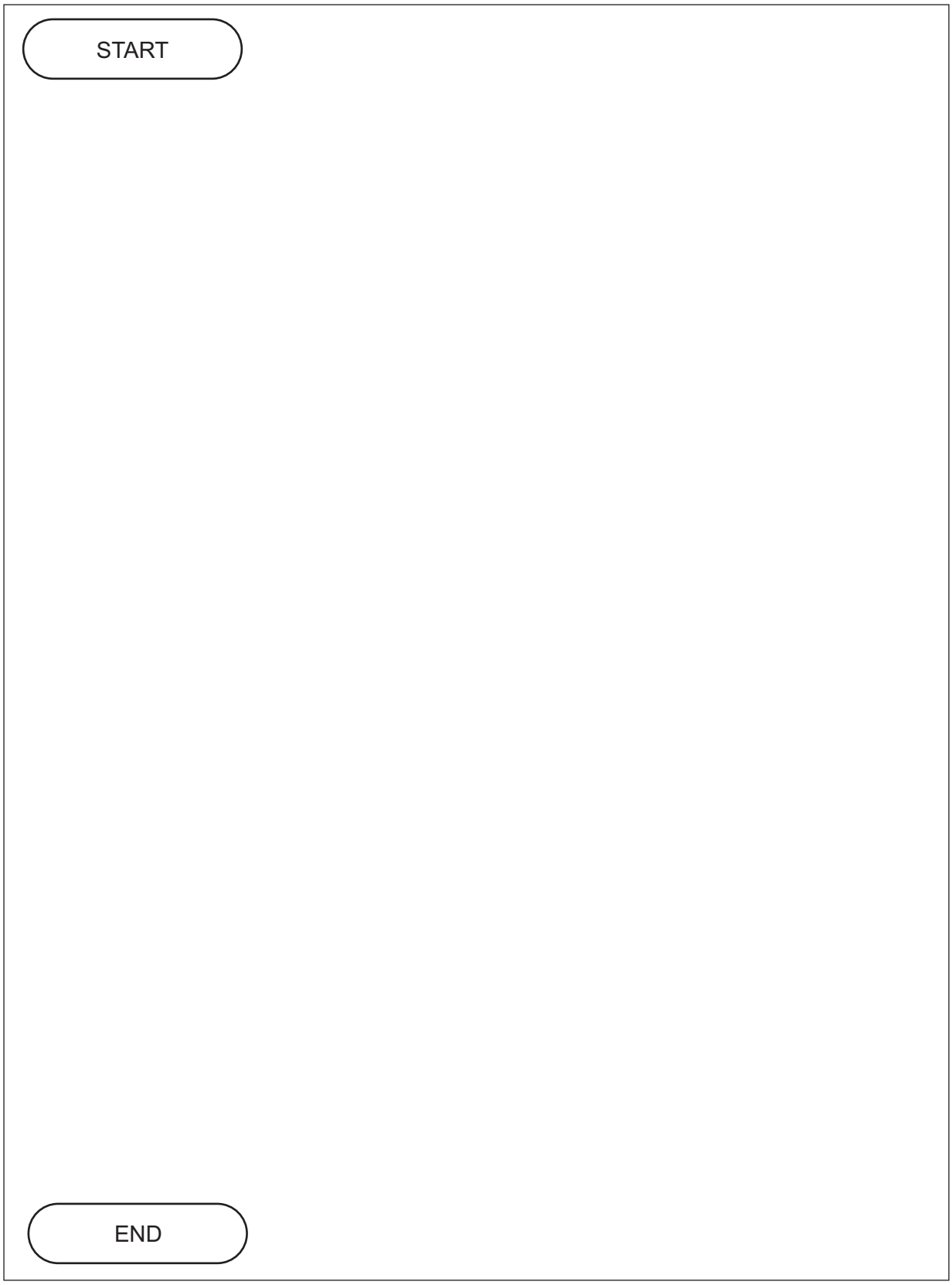
.....

[2]

2 `Data` is a 1D array of integers, containing 30 elements. All element values are unique.

(a) An algorithm will output the index of the element with the smallest value.

Draw a program flowchart to represent the algorithm.

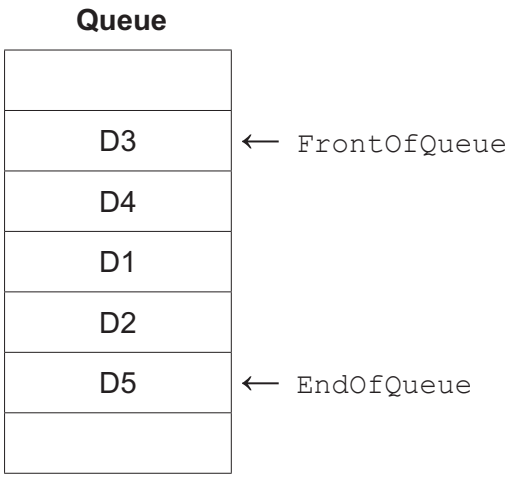


[5]

3 The diagram represents a queue Abstract Data Type (ADT).

The organisation of this queue may be summarised as follows:

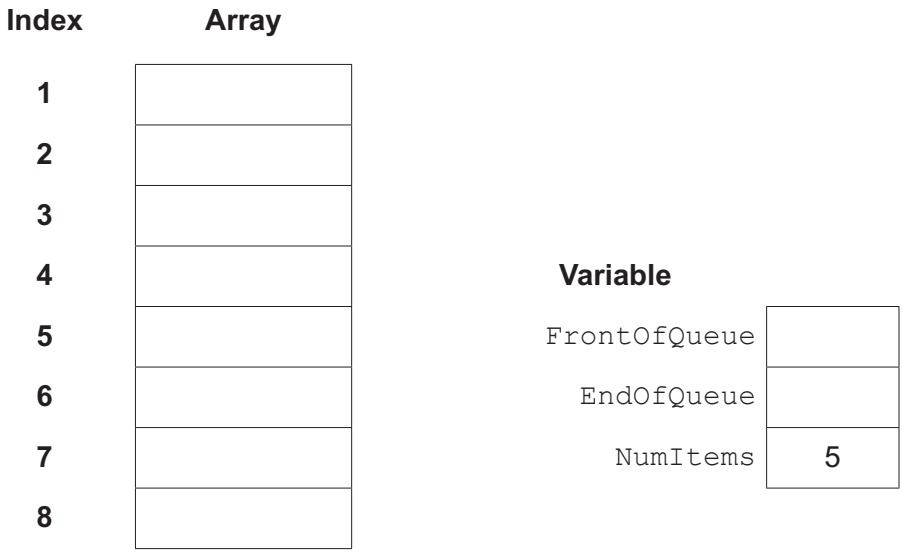
- The `FrontOfQueue` pointer points to the next data item to be removed.
- The `EndOfQueue` pointer points to the last data item added.



The queue is implemented using three variables and a 1D array of eight elements as shown. The variable `NumItems` stores the number of items in the queue.

The pointer variables store indices (index numbers) of the array.

(a) Complete the diagram to represent the state of the queue as shown above.



[3]

(b) A module `AddTo()` will add a value to the queue by manipulating the array and variables in part **(a)**.

The queue implementation is circular. When pointers reach the end of the queue, they will 'wrap around' to the beginning.

Before a value can be added to the queue, it is necessary to check the queue is not full.

The algorithm to add a value to the queue is expressed in six steps.

Complete the steps.

1. If `NumItems` ..... then jump to step 6.
2. Increment ..... .
3. If ..... then set `EndOfQueue` to ..... .
4. Increment ..... .
5. Set the ..... at the index stored in ..... to the ..... being added.
6. Stop.

[6]

4 A procedure `RandList()` will output a sequence of 25 random integers, where each integer is larger than the previous one.

(a) Write pseudocode for procedure `RandList()`.

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..... [6]

(b) Procedure `RandList()` is modified so that the random numbers are also written into a 1D array `Result`.

A new module is written to confirm that the numbers in the array are in ascending order.

This module contains an `IF` statement that performs a comparison between elements:

```
IF (Result[x + 1] = Result[x]) OR (Result[x] > Result[x + 1]) THEN
   Sequence ← FALSE
ENDIF
```

Write a simplified version of the conditional clause.

.....

.....

..... [1]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A shop sells car accessories. A customer order is created if an item cannot be supplied from current stock. A program is being developed to create and manage the customer orders.

(a) The following identifier table shows some of the data that will be stored for each order.

Complete the identifier table by adding meaningful variable names and appropriate data types.

Example value	Explanation	Variable name	Data type
`"Mr Khan"`	The name of the customer		
`3`	The number of items in the order		
`TRUE`	To indicate whether this is a new customer		
`15.75`	The deposit paid by the customer		

[4]

2 An algorithm will:

1. input a sequence of integer values, one at a time
2. ignore all values until the value 27 is input, then sum the remaining values in the sequence
3. stop summing values when the value 0 is input and then output the sum of the values.

(a) Draw a program flowchart to represent the algorithm.



[5]

2 **(a)** An algorithm is expressed as follows:

- input 100 numbers, one at a time
- keep a total of all numbers input that have a value between 30 and 70 inclusive and output this total after the last number has been input.

Outline, using stepwise refinement, the five steps for this algorithm which could be used to produce pseudocode.

Do **not** use pseudocode statements in your answer.

Step 1 .....

.....

Step 2 .....

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Step 3 .....

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Step 4 .....

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Step 5 .....

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[5]

(b) Sequence is one programming construct.

Identify **two other** programming constructs that will be required when the algorithm is converted into pseudocode.

Construct 1 .....

.....

Construct 2 .....

.....

[2]

8 A class of students are developing a program to send data between computers. Many computers are connected together to form a wired network. Serial ports are used to connect one computer to another.

Each computer:

- is assigned a unique three-digit ID
- has three ports, each identified by an integer value
- is connected to between one and three other computers.

Messages are sent between computers as a string of characters organised into fields as shown:

```
<STX><DestinationID><SourceID><Data><ETX>
```

Field number	Field name	Description
n/a	`STX`	a single character marking the start of the message (ASCII value 02)
1	`DestinationID`	three numeric characters that identify the destination computer
2	`SourceID`	three numeric characters that identify the source computer
3	`Data`	a variable length string containing the data being sent (Minimum length is 1 character)
n/a	`ETX`	a single character marking the end of the message (ASCII value 03)

For example, the following message contains the data "Hello Kevin" being sent from computer "101" to computer "232":

```
<STX>"232101Hello Kevin"<ETX>
```

Each computer will run a copy of the same program. Each program will contain a global variable, `MyID` of type string, that contains the unique ID of the computer in which the program is running.

The programmer has defined the first two program modules as follows:

Module	Description
`Transmit()` (already written)	• takes two parameters: ○ a string containing a message ○ an integer containing a port number • transmits the message using the given port
`SendFile()`	• takes three parameters: ○ a string containing a text file name ○ a string containing a Destination ID ○ an integer containing a Port number • transmits the file one line at a time • transmits a final message with data string "`****`"

(a) Write pseudocode for module `SendFile()`.

Assume:

- module `Transmit()` has already been written and is used to transmit a message
- the value of `MyID` may be used as `SourceID`
- the file specified contains no blank lines
- the file specified does not contain the line "`****`"

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..... [7]

(b) Module `SendFile()` is used to copy a file from one computer to another.

A module within the program running on the destination computer will receive the data and write it to a new file.

Explain why module `SendFile()` transmits the message with data string `"****"` after the last line of the file.

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.....

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..... [2]

(c) One of the text files to be sent contains several blank lines (lines that do not contain any text).

(i) Explain why this is a problem.

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.....

..... [2]

(ii) Explain how the message format could be changed to allow a blank line to be sent.

.....

.....

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..... [2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 The following pseudocode represents part of the algorithm for a program.

Line numbers are for reference only.

```
10   DECLARE Sheet4 : ARRAY[1:2, 1:50] OF INTEGER
…
100  FOR PCount ← 0 TO 49
101     Sheet4[1, PCount] ← 0
102     Sheet4[2, PCount] ← 47
103  NEXT PCount
```

(a) The pseudocode contains references to an array.

Complete the table by writing the answer for each row.

	Answer
The dimension of the array	
The name of the variable used as an array index	
The number of elements in the array	

[3]

(d) The pseudocode statements given in the following table are used in other parts of the algorithm.

Complete the table by placing **one or more** ticks (✓) in each row.

The first row has already been completed.

Pseudocode statement	**Input**	**Process**	**Output**
`INPUT MyChoice`	✓		
`OUTPUT FirstName & LastName`			
`WRITEFILE YourFile, TextLine`			
`READFILE MyFile, TextLine`			
`Result ← SQRT(NextNum)`			

[4]

2 A program stores a date of birth for a student using a variable, `MyDOB`, of type `DATE`.

(a) `MyDOB` has been assigned a valid value corresponding to Kevin's date of birth.

Complete the pseudocode statement to test whether Kevin was born on a Thursday.

`IF` ..... `THEN` [2]

(b) A function `CheckDate()` will take three integer parameters representing a day, month and year of a given date.

The function will validate the date of birth for a student that the parameters passed to it represent.
For a date to be valid, a student must be at least 18 in year 2020.

(i) Two of the parameter values can be checked without reference to the third parameter.

Describe these **two** checks.

Check 1 .....

.....

.....

Check 2 .....

.....

.....

.....
[2]

(ii) Several values of the parameter representing the day can only be checked completely by referring to the value of **one other** parameter.

Describe this check.

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..... [2]

3 A program processes data using a stack. The data is copied to a text file before the program ends.

(a) The following diagram shows the current state of the stack.

The operation of this stack may be summarised as follows:

- The `TopOfStack` pointer points to the last item added to the stack.
- The `BottomOfStack` pointer points to the first item on the stack.
- The stack grows upwards when items are added.

Stack		Pointer
Memory location	**Value**	
506		
505	WWW	← `TopOfStack`
504	YYY	
503	XXX	
502	ZZZ	
501	NNN	
500	PPP	← `BottomOfStack`

(i) An error will be generated if an attempt is made to POP a value when the stack is empty.

State the maximum number of consecutive POP operations that could be performed on the stack shown above **before** an error is generated.

..... [1]

(b) The data is copied to a text file before the program ends.

(i) State an advantage of writing the data from the stack to a text file before the program ends.

.....

..... [1]

(ii) A module `SaveStack()` will write the data from the stack to a text file.

Express an algorithm for `SaveStack()` as five steps that could be used to produce pseudocode.

Write the **five** steps.

Step 1 .....

.....

Step 2 .....

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Step 3 .....

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Step 4 .....

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Step 5 .....

.....

[5]

6 A procedure `Select()` will:

1. take two integer values as parameters representing start and end values where both values are greater than 9 and the end value is greater than the start value
2. output **each** integer value between the start and the end value (**not** including the start and end values), where the sum of the last two digits is 6, for example, 142.

(a) Write pseudocode for procedure `Select()`.

Parameter validation is **not** required.

..... [7]

(b) The check performed by procedure `Select()` on the last two digits is needed at several places in the program and will be implemented using a new function.

The new function `CheckNum()` will:

- allow the required sum to be specified (not just 6)
- check one number
- return an appropriate value.

Describe the function interface **and two** advantages of this modular approach.

Interface .....

.....

.....

.....

Advantage 1 .....

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Advantage 2 .....

.....

[4]

7 A school has a library system which allows students to borrow books for a length of time. Information relating to students and books is stored in text files. Student information includes name, home address, email address, date of birth, tutor and subject choices. Book information includes author, title, subject category, library location and the date that the book was borrowed.

A program helps the staff to manage the borrowing of books.

(a) A new module needs to be written to generate emails to send to students who have an overdue book. Students who are sent an email are prevented from borrowing any more books until the overdue book is returned.

The process of abstraction has been used when designing the new module.

(i) State the purpose of applying abstraction to this problem.

.....

..... [1]

(ii) Identify **one** item of information that is required and **one** item that is **not** required in the new module. Justify your choices.

Item required .....

Justification .....

.....

.....

Item not required .....

Justification .....

.....

.....

[2]

(iii) Identify **two** operations that would be required to process data when an overdue book is returned.

Operation 1 .....

.....

Operation 2 .....

.....

[2]

8 A computer shop assembles desktop computers, using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

```
<ItemNum><SupplierCode><Description>
```

Item information is as follows:

	Format	Comment
`ItemNum`	4 numeric characters	unique number for each item in the range “0001” to “5999” inclusive
`SupplierCode`	3 alphabetic characters	code to identify the supplier of the item
`Description`	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does not contain all possible values in the range.

The programmer has defined the first program module as follows:

Module	Description
`ChangeSupp()`	• called with two parameters `Code1` and `Code2` of type string that represent valid supplier codes • creates a new file `NewStock.txt` from the contents of the file `Stock.txt` where any reference to `Code1` is replaced by `Code2` • returns a count of the number of items that have had their supplier code changed

(a) Write pseudocode for module `ChangeSupp()`.

[8]

(c) The format of the output from module `Report_1()` from **part (b)** is changed. The number of items listed is moved to the top of the report as shown in the example:

```
Report for Supplier: DRG
Number of items listed: 3

Item        Description

1234        USB Printer Cable 3m
1273        32GB USB Flash Drive
1350        Mouse Mat 320 x 240mm
```

(i) Explain why this new layout would increase the complexity of the algorithm.

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..... [2]

(ii) The algorithm will be modified to produce the report in the new format. The modified algorithm will be implemented so that the file `Stock.txt` is only read once.

Describe the modified algorithm.

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..... [3]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program calculates the postal cost based on the weight of the item and its destination. Calculations occur at various points in the program and these result in the choice of several possible postal costs. The programmer has built these postal costs into the program.

For example, the postal cost of $3.75 is used in the following lines of pseudocode:

```
IF Weight < 250 AND ValidAddress = TRUE THEN
   ItemPostalCost ← 3.75    // set postal cost for item to $3.75
   ItemStatus ← "Valid"     // item can be sent
ENDIF
```

(a) **(i)** Identify a more appropriate way of representing the postal costs.

..... [1]

(b) The lines of pseudocode contain features that make them easier to understand.

State **three** of these features.

1 .....

2 .....

3 .....

[3]

2 A program stores a user's date of birth using a variable `MyDOB` of type `DATE`.

(a) Write a pseudocode statement, using a function from the **insert**, to assign the value corresponding to 17/11/2007 to `MyDOB`.

..... [1]

(c) The program will output the day of the week corresponding to `MyDOB`.

For example, given the date 22/06/2023, the program will output `"Thursday"`.

An algorithm is required. An array will be used to store the names of the days of the week.

Define the array **and** describe the algorithm in **four** steps.

Do **not** use pseudocode statements in your answer.

Array definition .....

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Step 1 .....

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Step 2 .....

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Step 3 .....

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Step 4 .....

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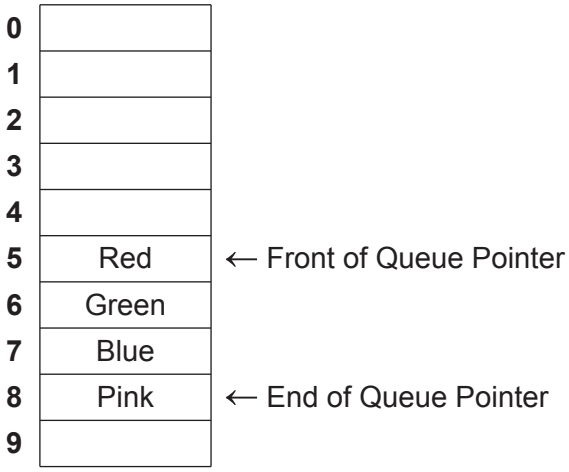
[6]

3 A program stores data in a text file. When data is read from the file, it is placed in a queue.

(a) The diagram below represents an Abstract Data Type (ADT) implementation of the queue. Each data item is stored in a separate location in the data structure. During initial design, the queue is limited to holding a maximum of 10 data items.

The operation of this queue may be summarised as follows:

- The Front of Queue Pointer points to the next data item to be removed.
- The End of Queue Pointer points to the last data item added.
- The queue is circular so that locations can be reused.



(i) Describe how the data items Orange and Yellow are added to the queue shown in the diagram.

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..... [4]

(b) The design of the queue is completed and the number of locations is increased.

A function `AddToQueue()` has been written. It takes a string as a parameter and adds this to the queue. The function will return `TRUE` if the string was added successfully.

A procedure `FileToQueue()` will add each line from the file to the queue. This procedure will end when all lines have been added or when the queue is full.

Describe the algorithm for procedure `FileToQueue()`.

Do **not** use pseudocode in your answer.

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..... [5]

7 A computer system for a shop stores information about each customer. The items of information include name and address (both postal and email) together with payment details and order history. The system also stores the product categories they are interested in and how they would like to be contacted.

(a) The shop wants to add a program module that will generate emails to be sent to customers who may be interested in receiving details of new products.

(i) State **three** items of information that the new module would need. Justify your choice in each case.

Information .....

Justification .....

.....

Information .....

Justification .....

.....

Information .....

Justification .....

.....

[3]

(ii) Identify **two** items of customer information that would **not** be required by the new module. Justify your choice in each case.

Information .....

Justification .....

.....

Information .....

Justification .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A programmer has written a program which includes the function `Calculate()`.
When the program is run, the function returns an unexpected value.

(a) Describe how a typical Integrated Development Environment (IDE) could be used to help debug the program to find the errors in the function `Calculate()`.

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..... [4]

(c) The program contains variables with values as follows:

Variable	Value
`Active`	`TRUE`
`Points`	`75`
`Exempt`	`FALSE`

(i) Complete the table by evaluating each expression.

	Expression	Evaluation
1	`(Points > 99) OR Active`	
2	`(Points MOD 2 = 0) OR Exempt`	
3	`(Points <= 75) AND (Active OR Exempt)`	
4	`(Active OR NOT Active) AND NOT Exempt`	

[2]

(ii) Write expression **4** from the table in part **(c)(i)** in its simplest form.

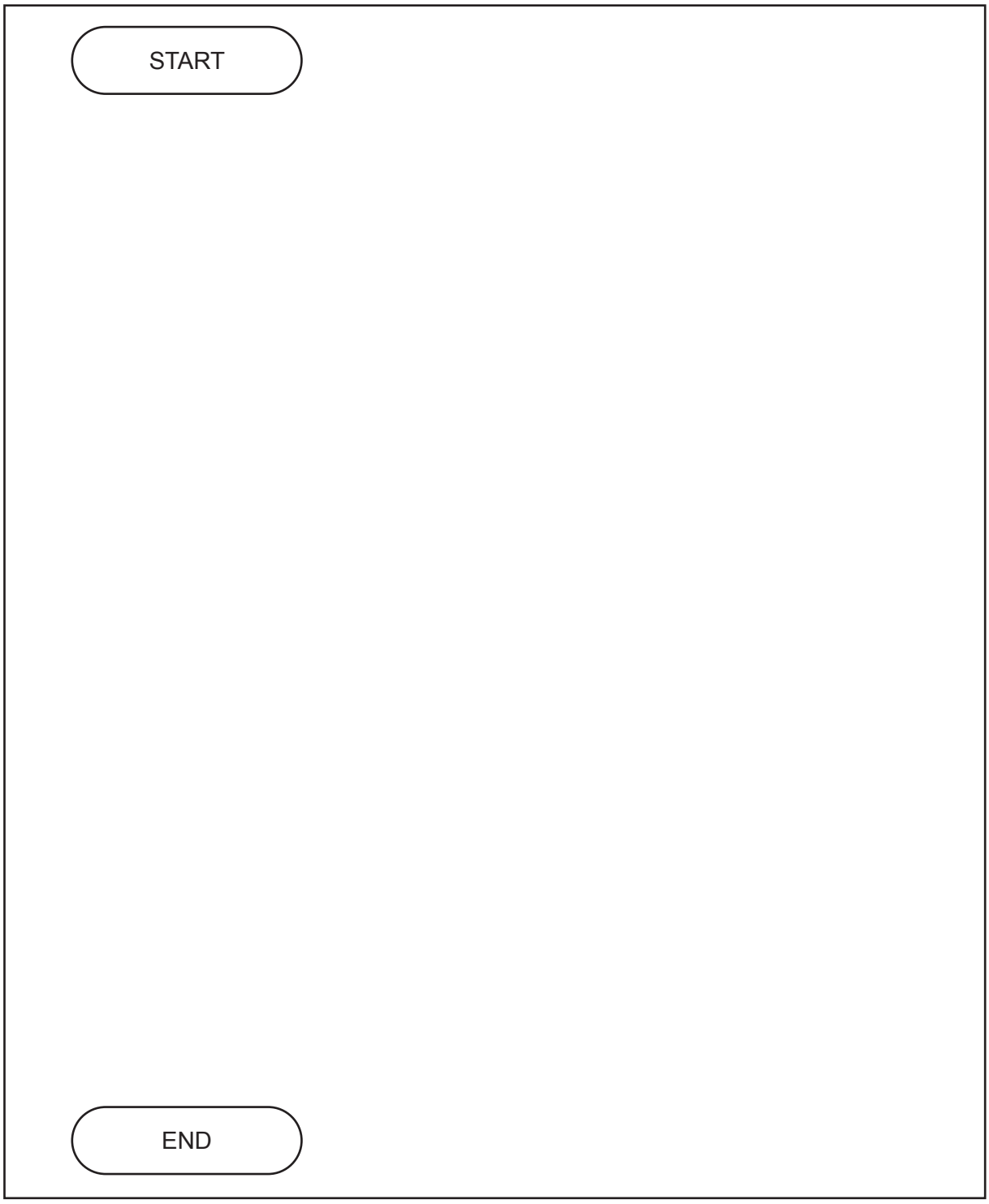
..... [1]

2 A program contains an algorithm to output a string of a specified length containing identical characters.

(a) The algorithm is described as follows:

1. prompt and input a character and store in `MyChar`
2. prompt and input an integer and store in `MyCount`
3. generate a string consisting of `MyChar` repeated `MyCount` times
4. output the string.

Draw a program flowchart to represent the algorithm.



[4]

3 Customers collect points every time they make a purchase at a store.

A program is used to manage the points system and the table lists some of the information stored for one customer.

Information	Data type required
Name	String
Number of points collected	Integer
Date of birth	Date

(a) (i) Identify a suitable structure for storing the information for one customer. Explain the advantage of using this structure.

Structure .....

Advantage .....

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[4]

(b) Customers receive points depending on the amount they spend. The number of points depends on the band that the amount falls into:

Band	Amount	Points
1	Less than $10	5 per whole dollar ($)
2	Between $10 and $100 inclusive	7 per whole dollar ($)
3	Over $100	10 per whole dollar ($)

For example, if the amount is $99.77, this amount is in band 2 and therefore the number of points is 7 × 99, which is 693 points.

The algorithm to calculate the points from a given amount is expressed as follows:

- work out the appropriate band
- calculate and output the number of points.

Apply the process of stepwise refinement to increase the detail of the algorithm. Structure your algorithm into a sequence of five steps that could be used to produce pseudocode.

Write the **five** steps.

1 .....

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2 .....

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4 .....

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5 .....

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[5]

7 A school has a computerised library system that allows students to borrow books for a fixed length of time. The system uses text files to store details of students, books and loans.

A new module is to be written which will generate emails to each student who has an overdue book.

(a) Decomposition will be used to break down the problem of designing the new module into sub-problems.

Identify **three** program modules that could be used in the design **and** describe their use.

Module 1 .....

Use .....

.....

.....

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Module 2 .....

Use .....

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.....

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Module 3 .....

Use .....

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[3]

2 A program is being designed for a smartphone to allow users to send money to the charity of their choice.

Decomposition will be used to break the problem down into sub-problems.

Identify **three** program modules that could be used in the design **and** describe their use.

Module 1 .....

Use .....

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.....

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Module 2 .....

Use .....

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.....

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Module 3 .....

Use .....

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[3]

3 An algorithm is needed to process a sequence of numbers. Numbers may be positive or negative and may be integer or decimal.

The algorithm will:

- prompt and input one number at a time until the value zero is input
- sum the negative numbers
- sum the positive numbers
- when zero is input, output the two sum values and then end.

Describe the algorithm needed. Do **not** include pseudocode statements in your answer.

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..... [5]

5 **(a)** A text string contains three data items concatenated as shown:

<StockID><Description><Cost>

Item lengths are:

Item	Length
StockID	5
Description	32
Cost	the remainder of the string

A procedure `Unpack()` takes four parameters of type string. One parameter is the original text string. The other three parameters are used to represent the three data items shown in the table and are assigned values within the procedure. These values will be used by the calling program after the procedure ends.

(i) Write pseudocode for the procedure `Unpack()`.

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(c) `Unpack()` is part of a program made up of several modules. During the design stage, it is important to follow good programming practice. One example of good practice is the use of meaningful identifier names.

Give the reason why this is good practice. Give **two other** examples of good practice.

Reason .....

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Example 1 .....

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Example 2 .....

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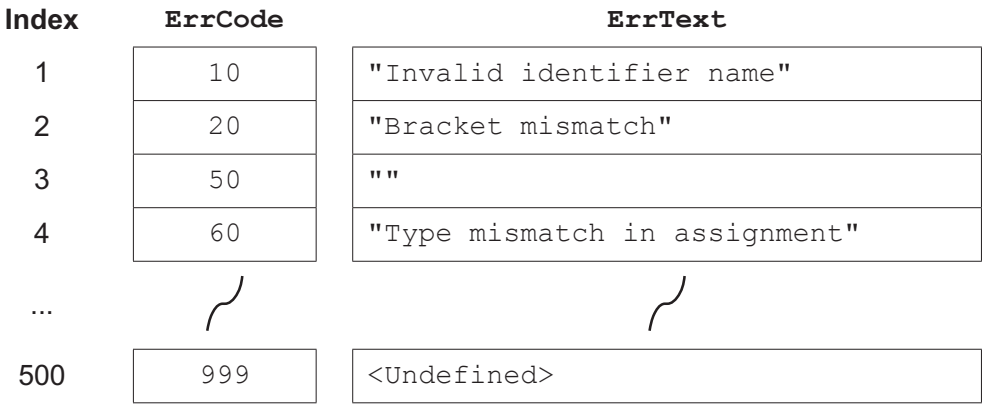
[3]

7 A teacher is designing a program to perform simple syntax checks on programs written by students.

Two global 1D arrays are used to store the syntax error data. Both arrays contain 500 elements.

- Array `ErrCode` contains integer values that represent an error number in the range 1 to 800.
- Array `ErrText` contains string values that represent an error description.

The following diagram shows an example of the arrays.



Note:

- There are less than 500 error codes so corresponding elements in both arrays may be unused. Unused elements in `ErrCode` have the value 999. These will occur at the end of the array. The value of unused elements in `ErrText` is undefined.
- Values in the `ErrCode` array are stored in ascending order but not all values may be present. For example, there may be no error code 31.
- Some error numbers are undefined. In these instances, the `ErrCode` array will contain a valid error number but the corresponding `ErrText` element will contain an empty string.

The teacher has defined one program module as follows:

Module	Description

`OutputRange()`	• Prompts for input of two error numbers
	• Outputs a list of error numbers between the two numbers input (inclusive) together with the corresponding error description
	• Outputs a warning message when the error description is missing as for error number 50 in the example
	• Outputs a suitable header and a final count of error numbers found
	
	Output based on the example array data above:
	
	`List of error numbers from 1 to 60`
	
	`10 : Invalid identifier name`
	`20 : Bracket mismatch`
	`50 : Error Text Missing`
	`60 : Type mismatch in assignment`
	
	`4 error numbers output`

(a) Write pseudocode for module `OutputRange()`. Assume that the two numbers input represent a valid error number range.

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Question 7 continues on the next page.

(b) (i) Two additional modules are defined:

Module	Description
`SortArrays()`	• Sorts the arrays into ascending order of `ErrCode`
`AddError()`	• Takes two parameters: ○ an error number as an integer ○ an error description as a string • Writes the error number and error description to the first unused element of the two arrays. Ensures the `ErrCode` array is still in ascending order • Returns the number of unused elements **after** the new error number has been added • Returns −1 if the new error number could not be added

Write pseudocode for the module `AddError()`. Assume that the error code is **not** already in the `ErrCode` array.

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(ii) A new Module `RemoveError()` will remove a given error number from the array.

Describe the algorithm that would be required. Do **not** include pseudocode statements in your answer.

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Refer to the **insert** for the list of pseudocode functions and operators.

1 **(a)** A programmer is developing an algorithm to solve a problem. Part of the algorithm would be appropriate to implement as a subroutine (a procedure or a function).

(i) State **two** reasons why the programmer may decide to use a subroutine.

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[2]

2 **(a)** An algorithm will process data from a test taken by a group of students. The algorithm will prompt and input the name and test mark for each of the 35 students.

The algorithm will add the names of all the students with a test mark of less than 20 to an existing text file `Support_List.txt`, which already contains data from other group tests.

(i) Describe the steps that the algorithm should perform.

Do **not** include pseudocode statements in your answer.

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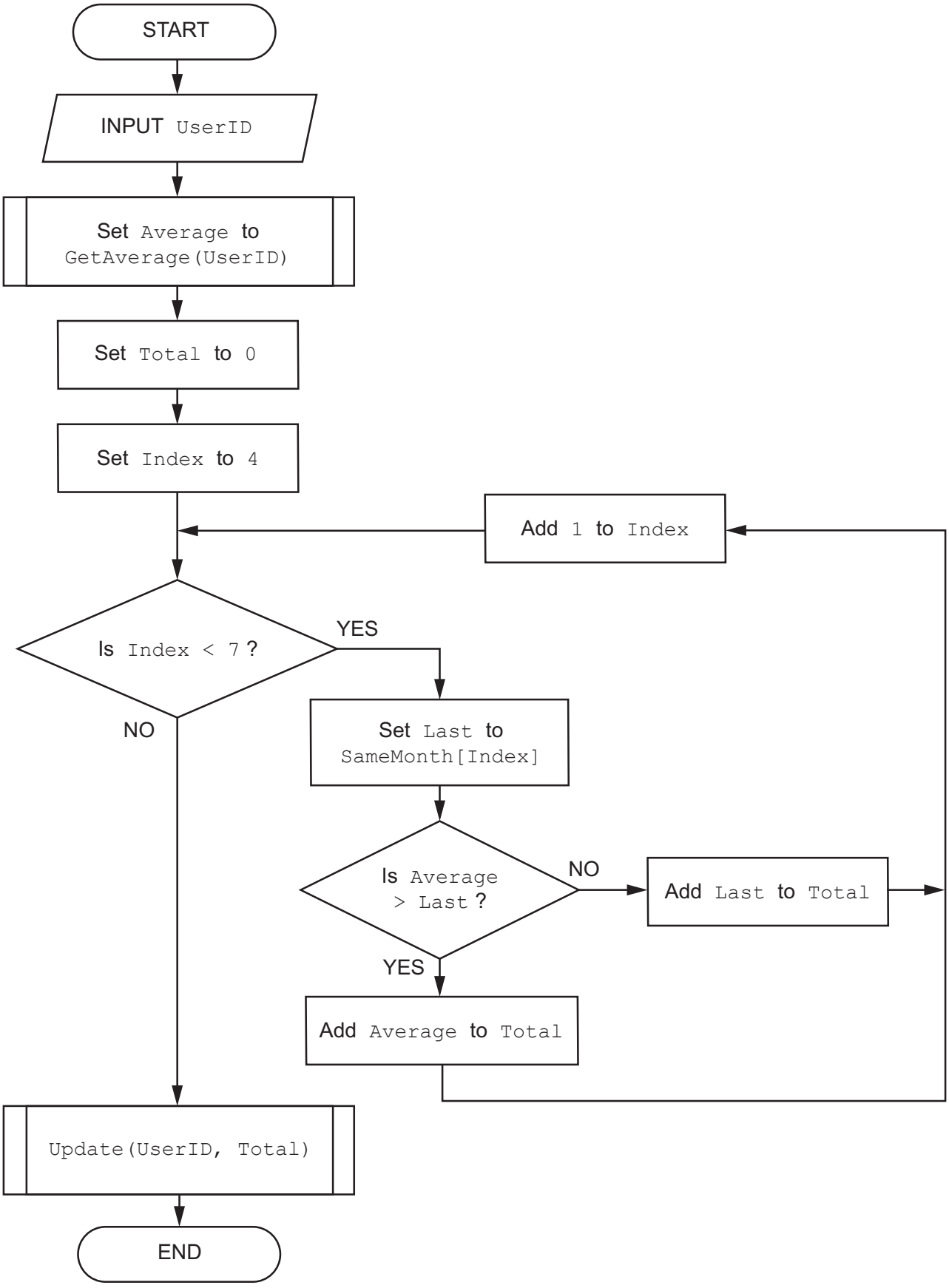
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4 The program flowchart represents a simple algorithm.



(a) Write the equivalent pseudocode for the algorithm represented by the flowchart.

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2 A system is being developed to help manage a car hire business. A customer may hire a car for a number of days.

An abstract model needs to be produced.

(a) Explain the process of abstraction **and** state **four** items of data that should be stored each time a car is hired.

Explanation .....

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Item 1 .....

Item 2 .....

Item 3 .....

Item 4 .....

[3]

(b) Identify **two** operations that would be required to process the car hire data.

Operation 1 .....

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Operation 2 .....

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[2]

3 A 1D array `Data` of type integer contains 200 elements. Each element has a unique value.

An algorithm is required to search for the largest value and output it.

Describe the steps that the algorithm should perform.

Do **not** include pseudocode statements in your answer.

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Refer to the **insert** for the list of pseudocode functions and operators.

1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.

(a) (i) State what her colleague means by "does not contain any syntax errors".

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(c) An airline wants to provide passengers with information about individual flights and allow them to book their flight using an online booking system.

(i) Tick (✓) **one** box in each row of the table to indicate whether each item of information would be essential for the customer when making the booking.

Information	Essential	Not essential
Departure time		
Flight number		
Departure airport		
Aircraft type		
Ticket price		
Number of seats in aircraft		

[3]

(ii) Identify the technique used to filter out information that is not essential when designing the booking system **and** state one benefit of this technique.

Technique .....

Benefit .....

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[2]

(iii) Identify **two additional** pieces of essential information that a passenger might need when booking a flight.

1 .....

2 .....

[2]

2 **(a)** An algorithm to sort a 1D array into ascending order is described as follows:

- move the largest value to the end
- keep repeating until the array is sorted.

Apply the process of stepwise refinement to this algorithm in order to produce a more detailed description.

Write the more detailed description using **structured English**. Your explanation of the algorithm should not include pseudocode statements.

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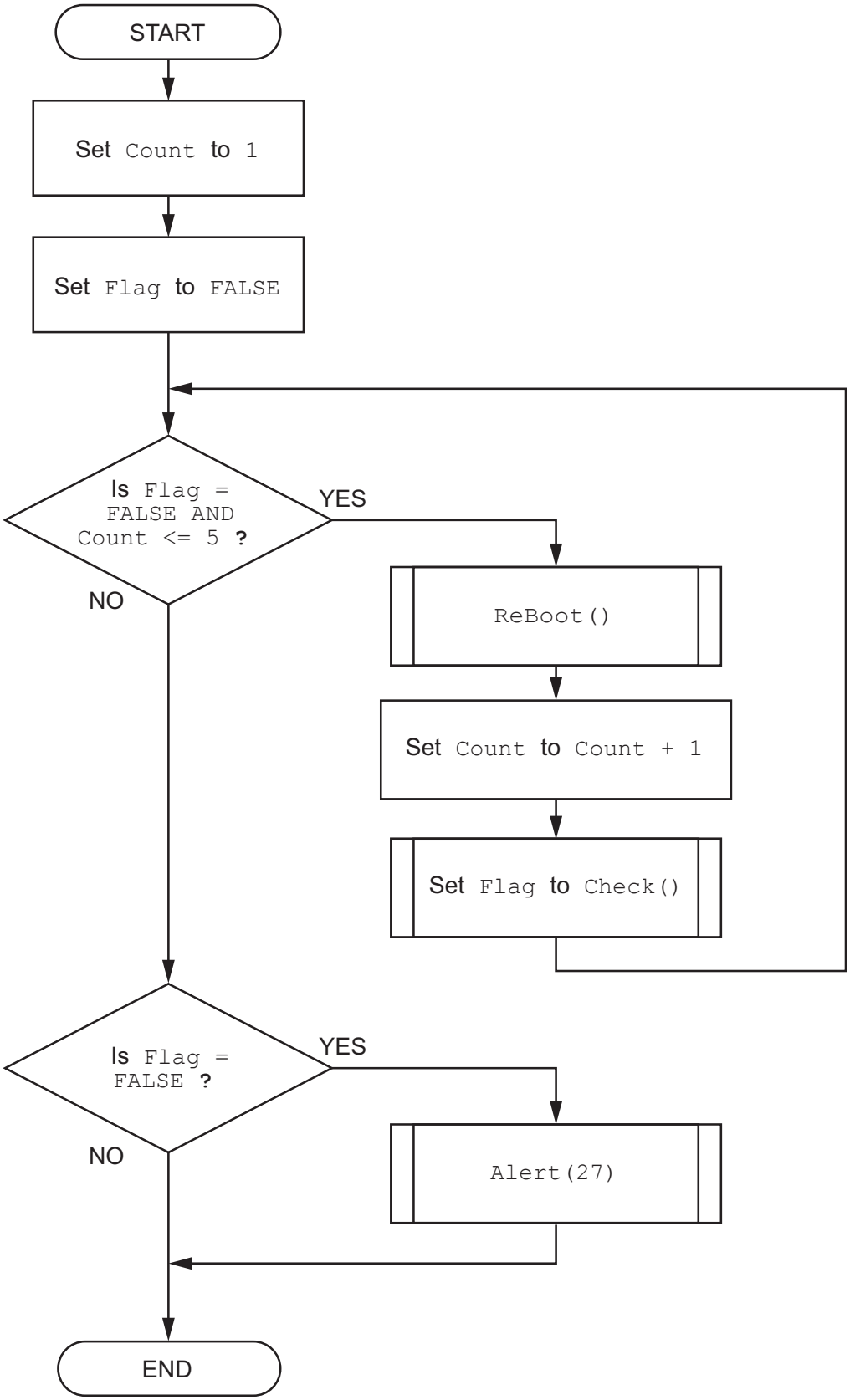
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(b) The program flowchart shown describes a simple algorithm.



Write pseudocode for the simple algorithm shown on page 6.

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3 **(a)** The diagram below represents a queue Abstract Data Type (ADT) that can hold a maximum of eight items.

The operation of this queue may be summarised as follows:

- The front of queue pointer points to the next item to be removed.
- The end of queue pointer points to the last item added.
- The queue is circular so that empty storage elements can be reused.

0	Frog	← Front of queue pointer
1	Cat	
2	Fish	
3	Elk	← End of queue pointer
4		
5		
6		
7		

(i) Describe how “Octopus” is added to the given queue.

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(c) The queue is implemented using a 1D array.

Describe the algorithm that should be used to modify the **end of queue pointer** when adding an item to the queue.

Your algorithm should detect any potential error conditions.

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Refer to the **insert** for the list of pseudocode functions and operators.

1 **(a)** A programmer applies decomposition to a problem that she has been asked to solve.

Describe decomposition.

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2 **(a)** An algorithm will:

1. input an integer value
2. jump to step 6 if the value is zero
3. sum and count the positive values
4. sum and count the negative values
5. repeat from step 1
6. output the two sum values and the two count values.

Draw a program flowchart on the following page to represent the algorithm.

Note that variable declarations are not required in program flowcharts.

[5]

4 The following is a procedure design in pseudocode.

Line numbers are given for reference only.

```
10  PROCEDURE Check(InString : STRING)
11     DECLARE Odds, Evens, Index : INTEGER
12
13     Odds ← 0
14     Evens ← 0
15     Index ← 1
16
17     WHILE Index <= LENGTH(InString)
18        IF STR_TO_NUM(MID(InString, Index, 1)) MOD 2 <> 0 THEN
19           Odds ← Odds + 1
20        ELSE
21           Evens ← Evens + 1
22        ENDIF
23        Index ← Index + 1
24     ENDWHILE
25
26     CALL Result(Odds, Evens)
27  ENDPROCEDURE
```

(a) Complete the following table by giving the answers, using the given pseudocode.

	Answer
A line number containing a variable being incremented	
The type of loop structure	
The number of functions used	
The number of parameters passed to `STR_TO_NUM()`	
The name of a procedure other than `Check()`	

[5]

(b) The pseudocode includes several features that make it easier to read and understand.

Identify **three** of these features.

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2 .....

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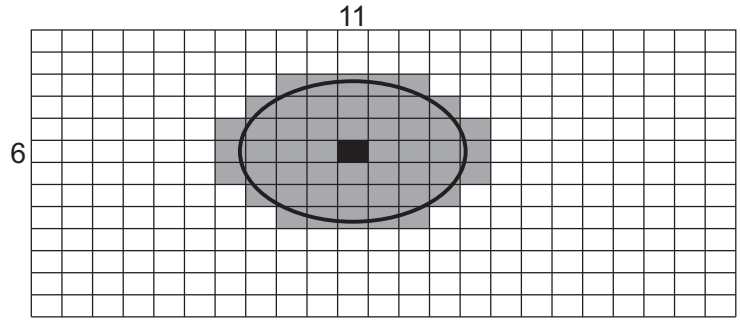
[3]

6 A mobile phone has a touchscreen. The screen is represented by a grid, divided into 800 rows and 1280 columns.

The grid is represented by a 2D array `Screen` of type `INTEGER`. An array element will be set to 0 unless the user touches that part of the screen.

Many array elements will set to 1 by a single touch of a finger or a stylus.

The following diagram shows a simplified touchscreen. The dark line represents a touch on the screen. All grid elements that are wholly or partly inside the outline will be set to 1. These elements are shaded.
The element shaded in black represents the centre point of the touch.



A program is needed to find the coordinates (the row and column) of the centre point. The centre point on the diagram shown is row 6, column 11.

Assume:

- the user may only touch one area at a time
- screen rotation does not affect the touchscreen.

The programmer has decided to use global values `CentreRow` and `CentreCol` as coordinate values for the centre point.

The programmer has started to define program modules as follows:

Module	Description
`FirstRowSet()`	• Searches for the first row that has an array element set to 1 • Returns the index of that row (1 is the first row) • Returns −1 if there are no elements set to 1
`LastRowSet()`	• Searches for the last row that has an array element set to 1 • Returns the index of that row • Returns −1 if there are no elements set to 1
`FirstColSet()`	• Searches for the first column that has an array element set to 1 • Returns the index of that column (1 is the first column) • Returns −1 if there are no elements set to 1
`LastColSet()`	• Searches for the last column that has an array element set to 1 • Returns the index of that column • Returns −1 if there are no elements set to 1

(a) Write efficient pseudocode for the module `FirstRowSet()`.

[7]

(c) The programmer decides to produce a **single** search module `FindSet()`, which will be able to perform each of the individual searches performed by the first four modules in the table.

(i) Outline the changes needed to convert one of the existing modules into this single module.

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(ii) Give one possible advantage **and** one possible disadvantage of combining the four searches into a single module.

Advantage .....

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Disadvantage .....

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[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.

(a) (i) State what her colleague means by "does not contain any syntax errors".

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..... [1]

(c) An airline wants to provide passengers with information about individual flights and allow them to book their flight using an online booking system.

(i) Tick (✓) **one** box in each row of the table to indicate whether each item of information would be essential for the customer when making the booking.

Information	Essential	Not essential
Departure time		
Flight number		
Departure airport		
Aircraft type		
Ticket price		
Number of seats in aircraft		

[3]

(ii) Identify the technique used to filter out information that is not essential when designing the booking system **and** state one benefit of this technique.

Technique .....

Benefit .....

.....

[2]

(iii) Identify **two additional** pieces of essential information that a passenger might need when booking a flight.

1 .....

2 .....

[2]

2 **(a)** An algorithm to sort a 1D array into ascending order is described as follows:

- move the largest value to the end
- keep repeating until the array is sorted.

Apply the process of stepwise refinement to this algorithm in order to produce a more detailed description.

Write the more detailed description using **structured English**. Your explanation of the algorithm should not include pseudocode statements.

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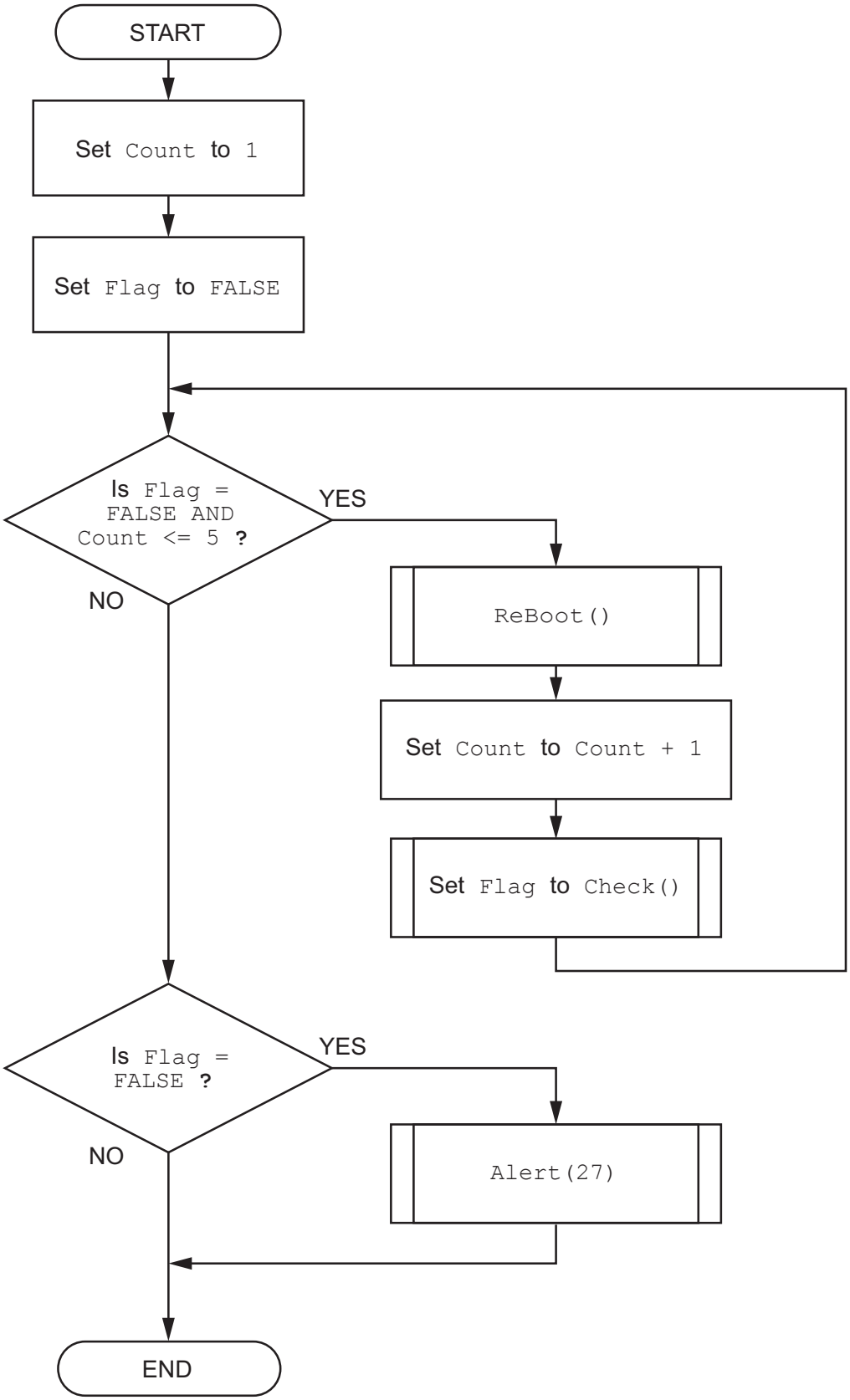
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(b) The program flowchart shown describes a simple algorithm.



Write pseudocode for the simple algorithm shown on page 6.

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Refer to the **insert** for the list of pseudocode functions and operators.

1 (a) A program is being developed to help manage the membership of a football club.

Complete the following identifier table.

Example value	Explanation	Variable name	Data type
`"Wong"`	The preferred name of the member joining the football club		
`FALSE`	A value to indicate whether an existing member of the club lives at the same address		
`19/02/1983`	When the member joined the football club		
`1345`	The number of points a member has earned. Members of the club earn points for different activities.		

[4]

2 **(a)** Four program modules form part of a program for a library.

A description of the relationship between the modules is summarised as follows:

Module name	Description
`UpdateLoan()`	• Calls either `LoanExtend()` or `LoanReturn()`
`LoanExtend()`	• Called with parameters `LoanID` and `BookID` • Calls `CheckReserve()` to see whether the book has been reserved for another library user • Returns `TRUE` if the loan has been extended, otherwise returns `FALSE`
`CheckReserve()`	• Called with `BookID` • Returns `TRUE` if the book has been reserved, otherwise returns `FALSE`
`LoanReturn()`	• Called with parameters `LoanID` and `BookID` • Returns a `REAL` (which is the value of the fine to be paid in the case of an overdue loan)

Draw a structure chart to show the relationship between the four modules and the parameters passed between them.

[5]

(c) A program will:

- input 50 unique integer values
- output the largest value
- output the average of the values **excluding** the largest value.

Draw a program flowchart to represent the algorithm.

Variable declarations are **not** required.

It is not necessary to check that each input value is unique.

[6]

7 A program is needed to take a string containing a full name and produce a new string of initials.

Some words in the full name will be ignored. For example, "the", "and", "of", "for" and "to" may all be ignored.

Each letter of the abbreviated string must be upper case.

For example:

Full name	Initials
Integrated Development Environment	IDE
The American Standard Code for Information Interchange	ASCII

The programmer has decided to use a global variable `FNString` of type `STRING` to store the full name.

It is assumed that:

- words in the full name string are separated by a single space character
- space characters will not occur at the beginning or the end of the full name string
- the full name string contains at least one word.

The programmer has started to define program modules as follows:

Module	Description
`GetStart()`	• Called with an `INTEGER` as a parameter, representing the number of a word in `FNString`. • Returns the character start position of that word in `FNString` or returns `-1` if that word does not exist • For example: if `FNString` contains the string `"hot and cold"`, `GetStart(3)` returns `9`
`GetWord()`	• Called with a parameter representing the position of the first character of a word in `FNString` • Returns the word from `FNString` • For example: if `FNString` contains the string `"hot and cold"`, `GetWord(9)` returns `"cold"`

(a) Write pseudocode for the module `GetStart()`.

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(b) The programmer has decided to use a global ten-element 1D array `IgnoreList` of type `STRING` to store the ignored words. Unused elements contain the empty string (`""`) and may occur anywhere in the array.

A new module `AddWord()` is needed as follows:

Module	Description
`AddWord()`	• Called with a parameter representing a word • Stores the word in an unused element of the `IgnoreList` array and returns `TRUE` • Returns `FALSE` if the array was already full or if the word was already in the array

Write a detailed description of the algorithm for `AddWord()`. Do **not** include pseudocode statements in your answer.

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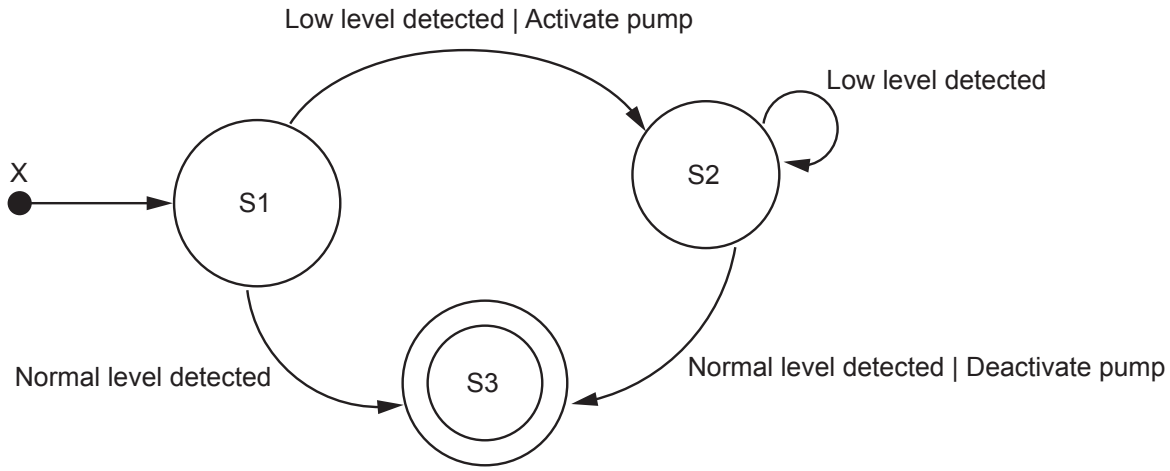
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2 **(a)** Examine the following state-transition diagram.



(i) Complete the table with reference to the diagram.

	Answer
The number of transitions that result in a different state	
The number of transitions with associated outputs	
The label that should replace 'X'	
The final or halting state	

[4]

(b) A system is being developed to help manage book loans in a library.

Registered users may borrow books from the library for a period of time.

(i) State **three** items of data that must be stored for each loan.

1 .....

2 .....

3 .....

[2]

(ii) State **one** item of data that will be required in the library system but does not need to be stored for each loan.

..... [1]

(iii) One operation that manipulates the data stored for each loan, would produce a list of all overdue books.

Identify **two other** operations.

Operation 1 .....

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Operation 2 .....

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[2]

4 A teacher uses a paper-based system to store marks for a class test. The teacher requires a program to assign grades based on these results.

The program will output the grades together with the average mark.

Write a detailed description of the algorithm that will be needed.

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Draw a structure chart to show the relationship between the four modules and the parameters passed between them.

[5]

(c) A program will:

- input 50 unique integer values
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(a) Write pseudocode for the module `GetStart()`.

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..... [7]

(b) The programmer has decided to use a global ten-element 1D array `IgnoreList` of type `STRING` to store the ignored words. Unused elements contain the empty string (`""`) and may occur anywhere in the array.

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